Advait Nair

832-493-4529 advaitnair.work@gmail.com linkedin.com/in/advaitnair10 U.S. Citizen

Professional Summary

First-year Civil Engineering student at UT Austin. Mechanical design experience across CAD modeling, engineering drawings, and hardware delivery on collaborative project teams, plus undergraduate research experience.

Education

UT Austin Austin, TX
B.S. in Civil Engineering Expected May 2030
Relevant coursework: Civil Engineering Systems; Sequences, Series, and Multivariable Calculus; Principles of Chemistry I

Engineering Experience

Longhorn Racing Electric — UT Austin Formula SAE Electric Austin, TX
Body Team Member Sep. 2026 – Present

- Supporting the design and build of a torsional-rigidity test jig used to measure chassis torsional stiffness, applying beam-deflection and first-principles structural analysis for the team's Formula SAE Electric race car
- Developed a Python sizing calculator that derives fixture and stand heights from applied load, chassis rotation, beam deflection, and floor-clearance constraints. The tool accelerated the calculation phase by nearly one month compared with the previous year's workflow

ShorelineAI Remote
Engineering Intern May 2024 – Aug. 2024

- Reverse engineered a wind-direction sensor in Onshape from photographs and physical measurements, creating a CAD model where no source model existed
- Designed a 3D-printable, bolt-on mount for an existing steel crossarm that prevents rotation without drilling; selected ASA material from first-principles assessment of anticipated field conditions and delivered the CAD assembly and engineering drawings to the hardware team

Korgel Group, UT Austin Austin, TX
CAD Design Contributor Oct. 2026 – Present

- Developing precise CAD models for shadow masks that define selective gold-deposition patterns on glass substrates during evaporation for solar-cell fabrication

Projects

NitrogenMAPS — Precision Fertilization Concept *Engineering Design Project, TSA*
4th of 45+ teams at the Texas TSA State Conference

- Co-developed an automated soil-nitrogen monitoring and fertilizer-delivery concept that releases fertilizer when measured nitrogen falls below a target threshold
- Benchmarked the concept against alternative approaches and proposed validation across component, controlled-field, and multi-season trials

Skills

Design & CAD: SolidWorks, Onshape, AutoCAD
Programming & Data: Python, R, Excel, HTML/CSS
Hands-on Fabrication: Basic TIG welding, 3D printing

Leadership & Activities

Technology Student Association - Chapter President *2022 – 2026*

- Led 120+ members through 90+ club meetings and managed up to $13,000 in chapter funds; placed 3rd in Debating Technical Issues at the International Conference

Seven Lakes Speech and Debate - Extemporaneous Speaking Captain *2022 – 2026*

- State finalist (4th in Impromptu of ∼1,000 competitors); national qualifier; 40th in Texas Congressional Debate; NSDA Premier Distinction (top 3% of members)